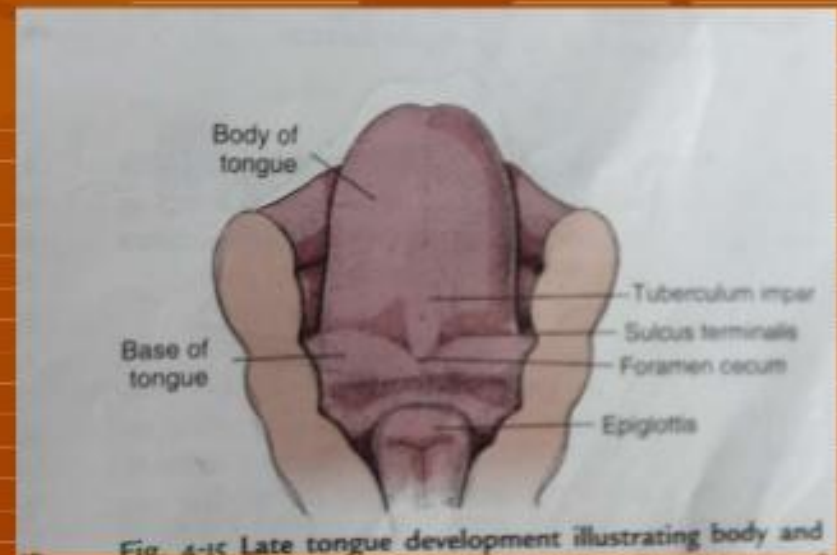


Development of **TONGUE**

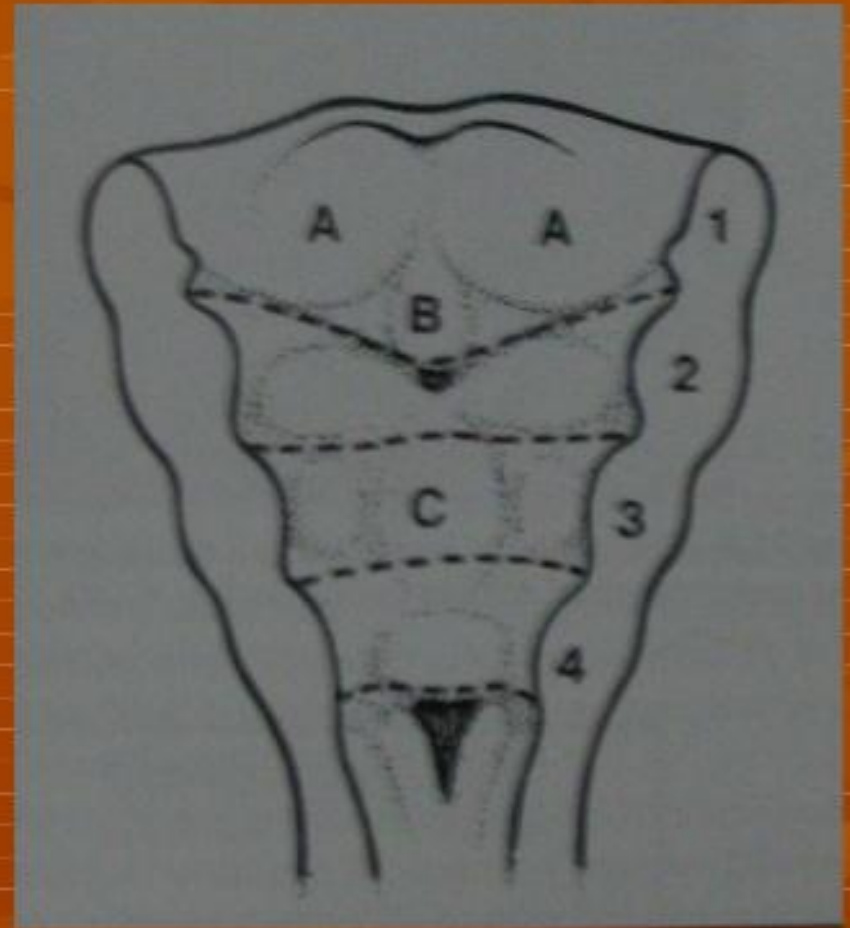
Parts of the tongue

- The first pharyngeal arch tissues forms the anterior (movable) body of the tongue
- The second & third arches form the posterior (immovable) body & base .

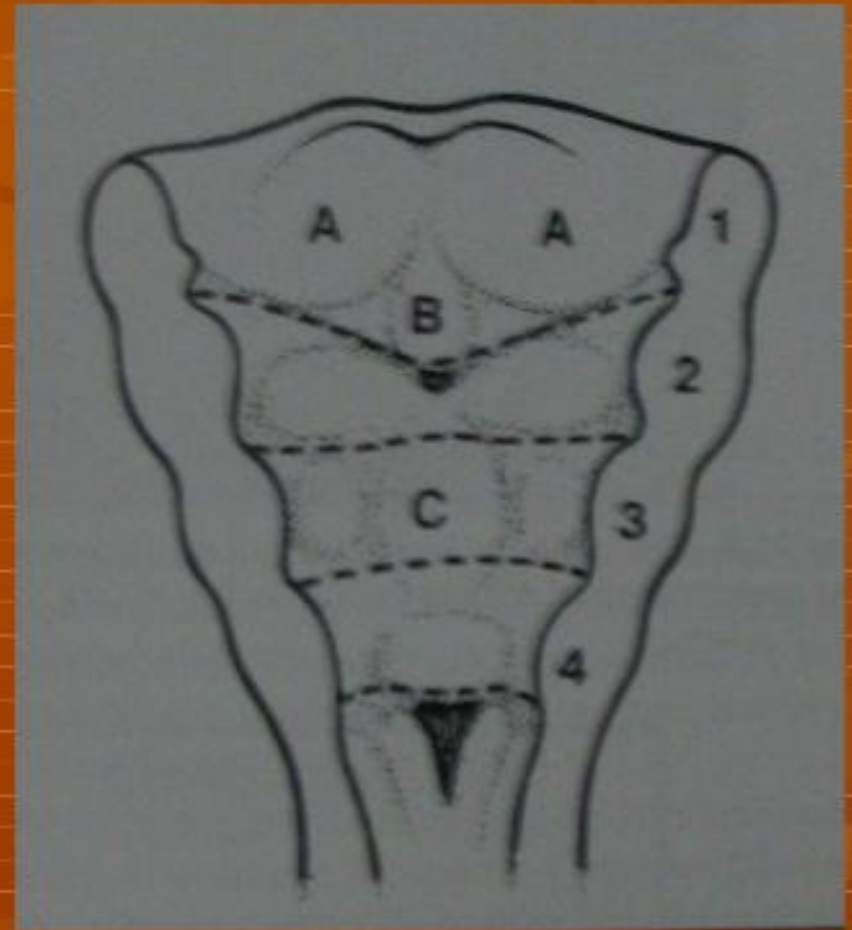


Formation of the Tongue

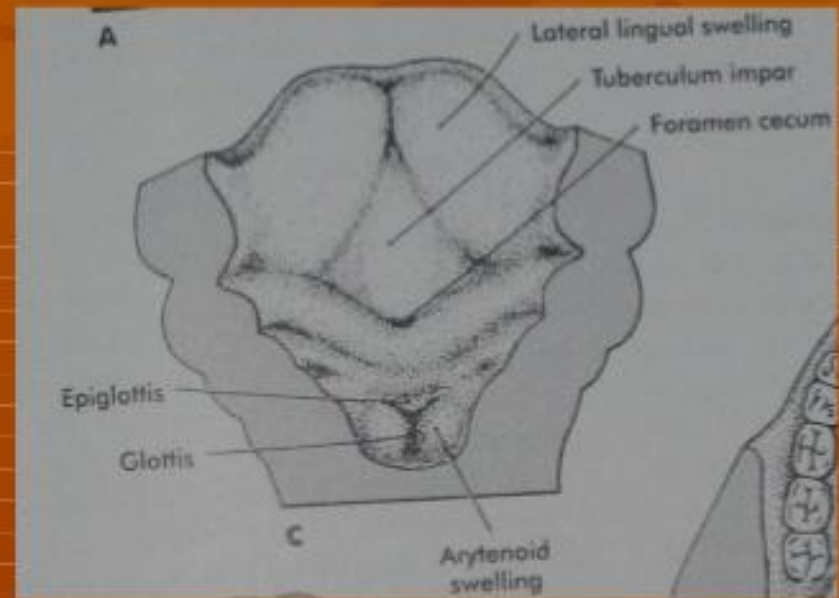
- The tongue begins to develop at about 4 weeks
- Local proliferation of the mesenchyme then gives rise to a number of swellings in the floor of the mouth



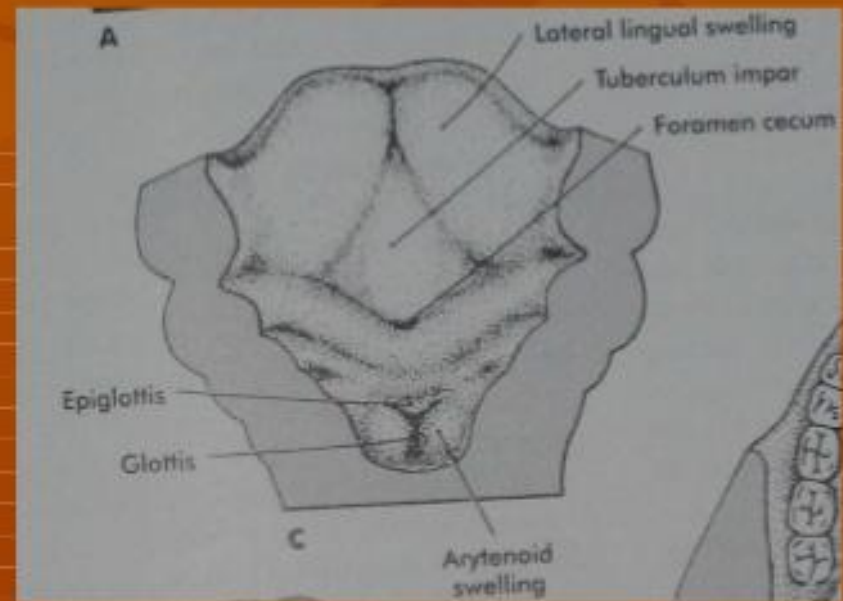
- First, a swelling (the **tuberculum impar**) arises in the midline in the mandibular process and is flanked by two other bulges, the **lingual swellings**



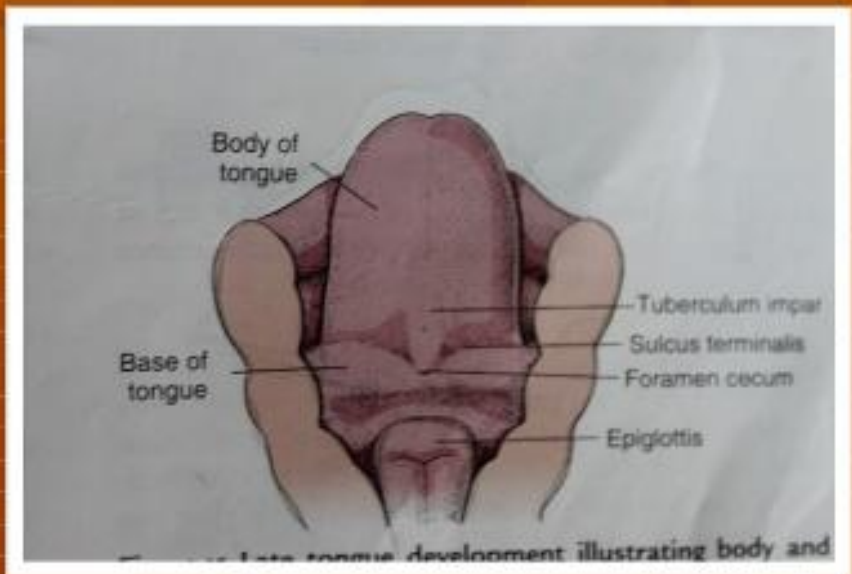
- Very quickly these lateral lingual swellings enlarge and merge with each other and the tuberculum impar to form a large mass from which the mucous membrane of the anterior two thirds of the tongue is formed



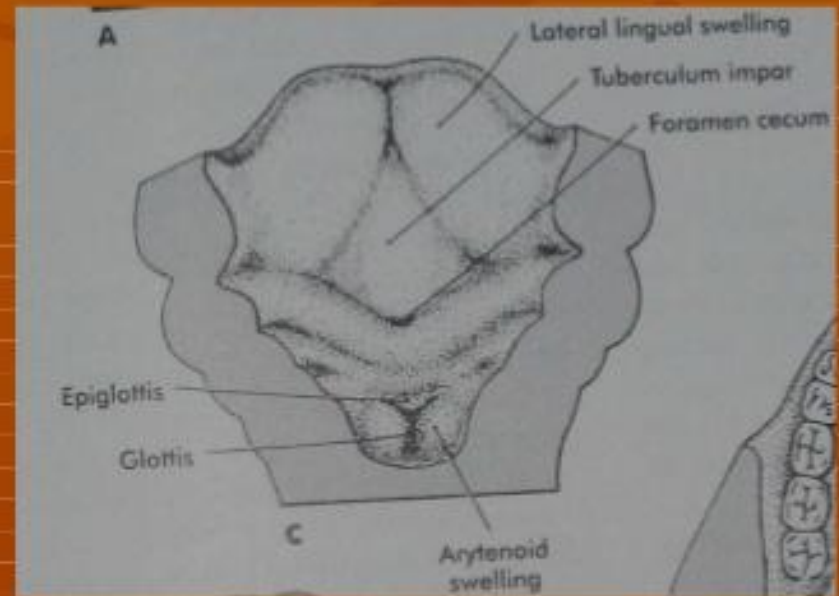
- The root of the tongue arises from the **hypobranchial eminence**, a large midline swelling developed from the mesenchyme of the third arch



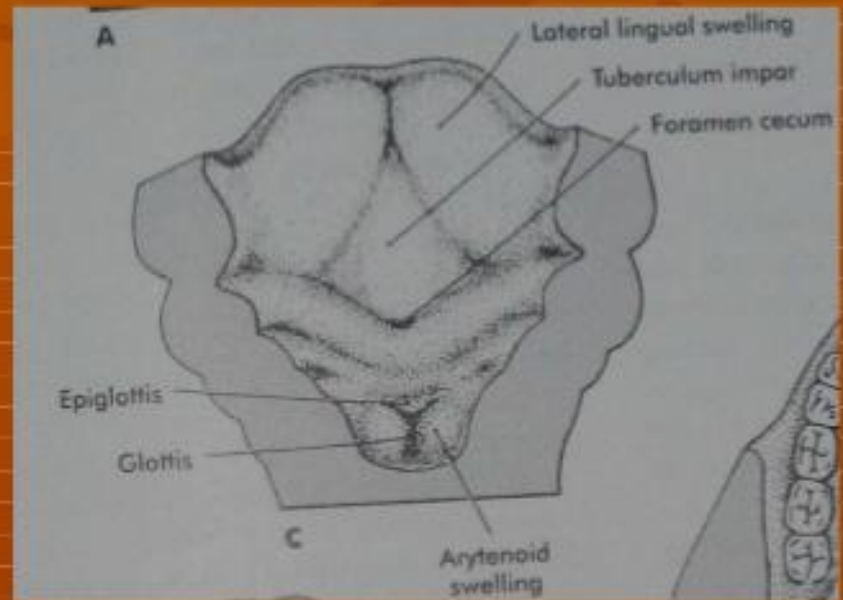
- Hypobranchial eminence is the primordial of the epiglottis (thin plate of cartilage in front of the glottis that protects it) during swallowing



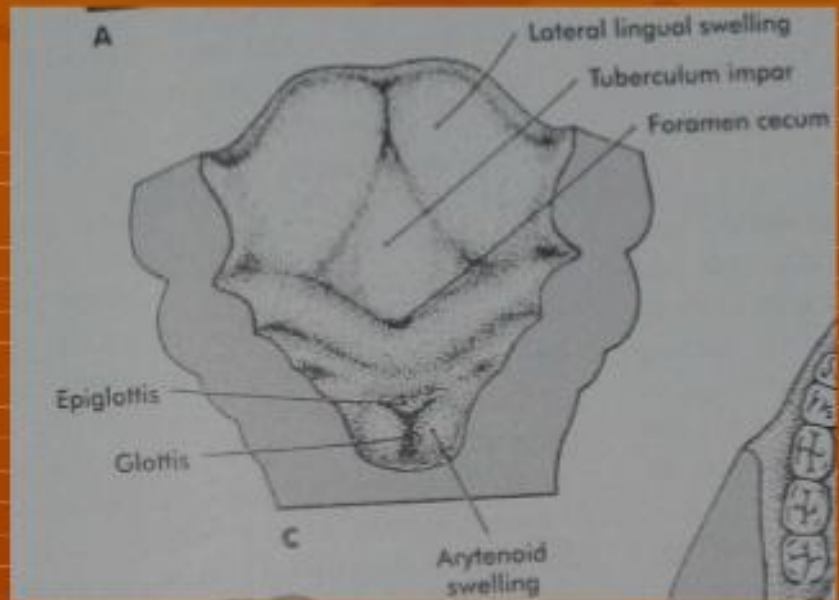
- The mesenchyme of the third arch rapidly overgrows that of the second arch, which is thereby excluded from further involvement in the development of the tongue



- The **hypobranchial eminence** gives rise to the mucous covering the root, or posterior third of the tongue



- Some authorities divide the **hypobranchial eminence** into an anterior copula (which gives origin to the mucosa covering the root of the tongue) and a **hypobranchial eminence** (which gives rise to the epiglottis)



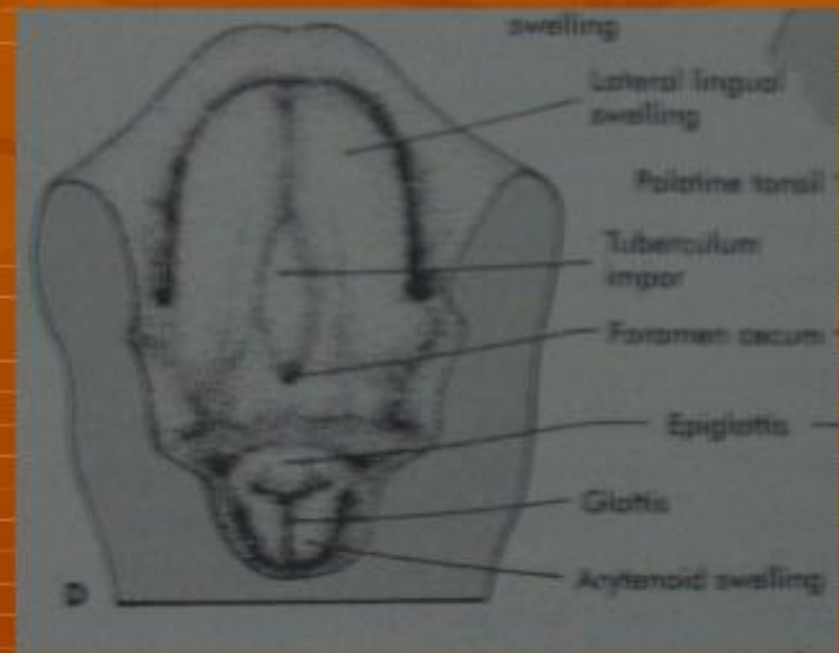
- The tongue separates from the floor of the mouth by a down growth of ectoderm around its periphery, which subsequently degenerates to form the lingual sulcus and gives the tongue mobility

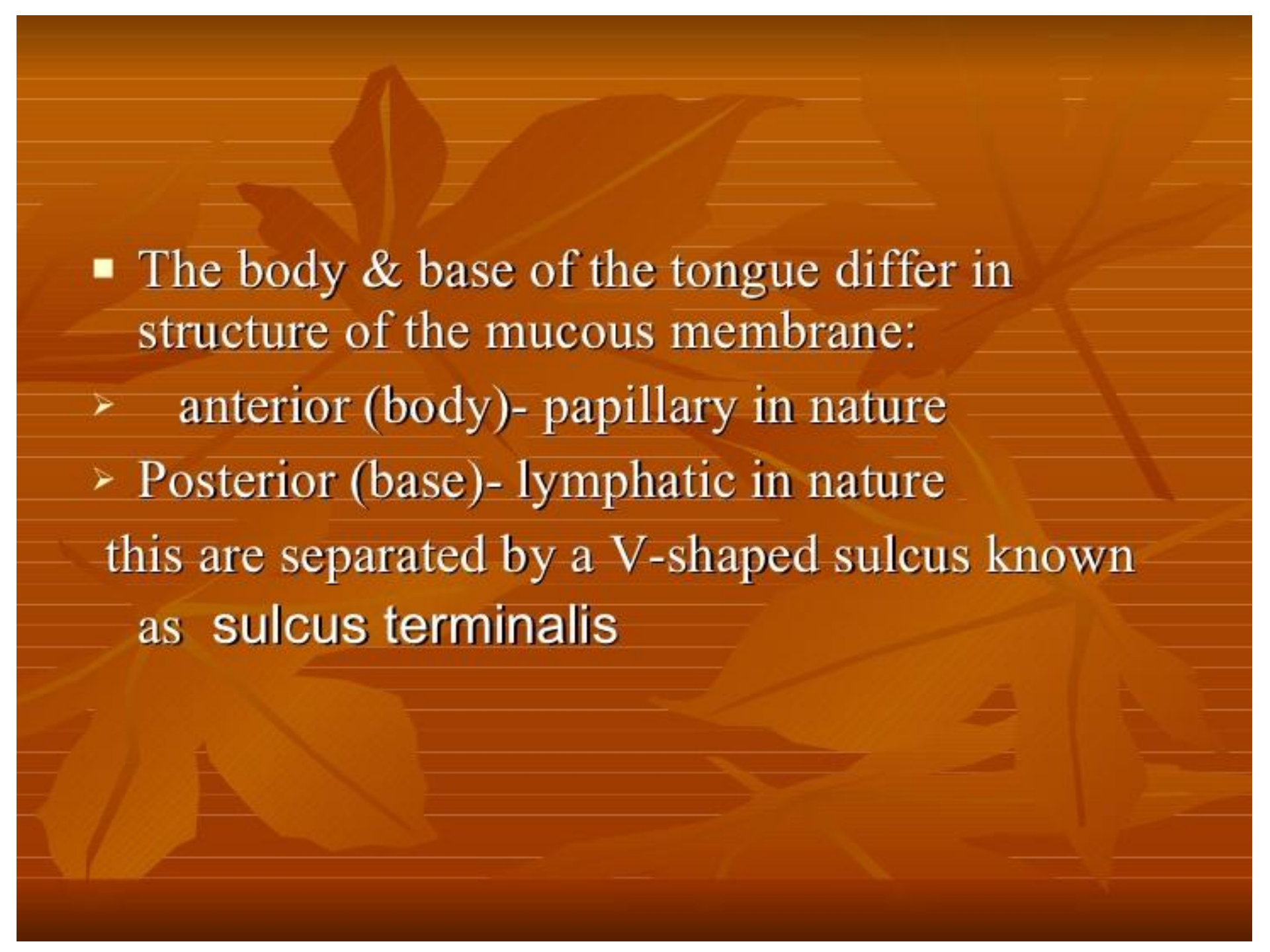


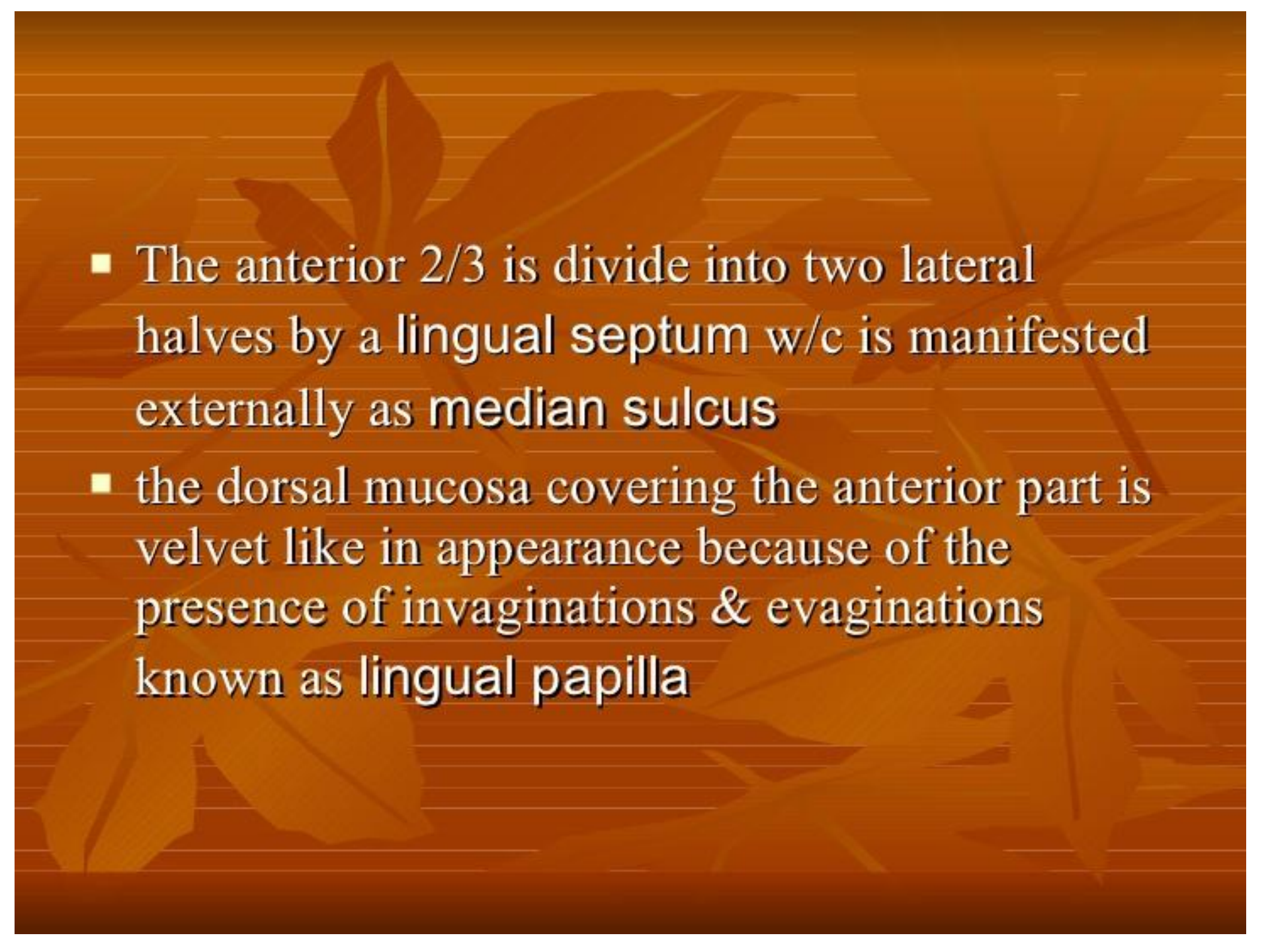
- The **muscles** of the tongue have a different origin: they arise from the **occipital somites**, which have migrated forward into the tongue area, carrying with them their supply, the twelfth cranial (**hypoglossal**) nerve



- This unusual development of the tongue explains its innervation
- Since the mucosa of the anterior two thirds of the tongue is derived from the first arch, it is supplied by the fifth cranial nerve (trigeminal nerve), whereas the mucosa of the posterior third of the tongue, derived from the third arch, is supplied by the ninth cranial nerve (glossopharyngeal nerve)



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- The body & base of the tongue differ in structure of the mucous membrane:
 - anterior (body)- papillary in nature
 - Posterior (base)- lymphatic in nature
- this are separated by a V-shaped sulcus known as **sulcus terminalis**

- 
- The anterior 2/3 is divide into two lateral halves by a lingual septum w/c is manifested externally as median sulcus
 - the dorsal mucosa covering the anterior part is velvet like in appearance because of the presence of invaginations & evaginations known as lingual papilla

Thank You!!